ECE 657D: Research Project

Effect of Model Quantization on Algorithmic Recourse

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Abstract

As algorithmic recourse advances, ensuring its robustness has emerged as a critical requirement for its practical value. Predictive models inevitably evolve due to retraining, deployment updates, or optimization constraints. While current recourse methods commonly rely on counterfactual explanations, generated by finding the minimal perturbation required to cross a model's decision boundary, little is known about how these explanations behave when the underlying model undergoes quantization. In this study, we investigate the effects of post-training quantization (PTQ) and quantization-aware training (QAT) on the validity of recourse under model updates. Our findings highlight the volatility of PTQ when seeking robust recourse and demonstrate that QAT yields models that better preserve the original model's predictive boundaries, thereby maintaining higher recourse validity. [1]

1 Introduction

AI-assisted decision-making is not a new concept, but the rapid advancement of AI development and its adoption in high-stakes industries, such as banking and healthcare, necessitates a deeper understanding of these systems' decision-making processes. One core aspect of explainable AI is the right to an explanation [3], which demands that any user affected by an AI-powered decision process has the right to a *meaningful* explanation of why that decision was made. Consequently, counterfactuals have emerged as a popular method to offer the meaningful explanations users desire, allowing them to understand the changes required to gain a favorable outcome. Counterfactuals provide a perturbed version of the initial factual instance, assuring the user that if they apply the actions required to reach the counterfactual state, they will be guaranteed a favorable outcome.

Preliminary work by Wachter et al. [12] offers an iterative methodology that utilizes gradient descent. The objective function aims to flip the factual prediction by perturbing the features while simultaneously minimizing the cost of the counterfactual. This cost is often measured as an l_p-based distance between the factual and counterfactual vectors, typically l_1. Due to this cost minimization, the resulting counterfactual often barely crosses the model's decision boundary, making it particularly fragile to slight shifts in that boundary. It is entirely reasonable to assume the boundary of a production model will shift as regular data updates occur. To address these concerns, algorithms for *robust recourse* [11] [9] have been developed to mitigate recourse fragility in the face of various potential changes [6] a model or factual instance may undergo.

Outside of core model updates, which are frequently studied due to their relevance and frequency, a less-discussed form of change is model quantization. Quantization is generally associated with large language models, particularly for local or edge deployments where memory footprint is a critical usability factor. However, its effects on decision boundaries apply broadly. There are two common

[1] I acknowledge the use of AI for the purposes of formatting and grammer. All ideas presented are my own.

techniques for model quantization, each offering different advantages and disadvantages. Our goal in this experiment is to explore the effects of both Post-Training Quantization (PTQ) and Quantization-Aware Training (QAT) on the validity of recourse, and to compare how robust these counterfactuals remain on future trained models.

2 Related Work

To the best of our knowledge, there is no direct prior work operating at the intersection of these specific topics. However, the separate fields of quantization and robust recourse have been extensively researched. For quantization, our primary foundation is Nagel et al. [8]. The core contribution of their white paper is the categorization of quantization paradigms, distinguishing specifically between simulated quantization (used primarily for theoretical evaluation of precision loss) and integer-only quantization (strictly required for realizing actual latency and power improvements on edge hardware). We apply these principles in our experimentation with both PTQ and QAT models. For algorithmic recourse, we refer primarily to three core works. The first is the foundational work by Wachter et al. [12], which uses gradient descent-based optimization to find the least-cost counterfactual. Subsequent work by Upadhyay et al. [11] suggested the move toward robust recourse, specifically addressing data shifts. They achieve this through a mini-max formulation of Wachter's method, accounting for worst-case perturbations applied to a model's weights within a defined l_2 radius δ. Finally, we incorporate the method by Pawelczyk et al. [9], which takes a probabilistic approach to offer robustness to perturbations in the application of the recourse itself. They achieve this by formulating a closed-form probabilistic measure of recourse invalidation, which can be tuned to suggest recourse that is more or less likely to become invalid based on user implementation. Because our goal is experimental and focused on the effects of quantization, we avoid diverging too deeply into the mathematical structure of the recourse generation methods themselves.

3 Method

We study the robustness of algorithmic recourse under model quantization and data shift. We aim to answer two central questions: 1) whether counterfactuals generated for a source non-quantized model remain valid when that same model is quantized, and 2) whether counterfactuals found using quantized models remain robust on future non-quantized models. We use a binary multilayer perceptron classifier with two hidden layers and a sigmoid output, trained using Adam and binary cross-entropy loss. The implementation defines both a standard FP32 model and a quantization-ready variant equipped with quant/dequant stubs and fused linear-ReLU blocks for quantization-aware training (QAT) in PyTorch. Experiments are run on CPUs to ensure quantization compatibility and reproducibility. The code is available here.

For each dataset, preprocessing includes stratified train/test splitting, standardization of numerical features, and one-hot encoding of categorical features where necessary. The global decision threshold is fixed at 0.5. We evaluate four different model variations: an FP32 current model, an INT8 PTQ version of the current model, an INT8 QAT version of the current model, and a FP32 future model. Recourse generation is performed exclusively on negatively classified test points under the source model.

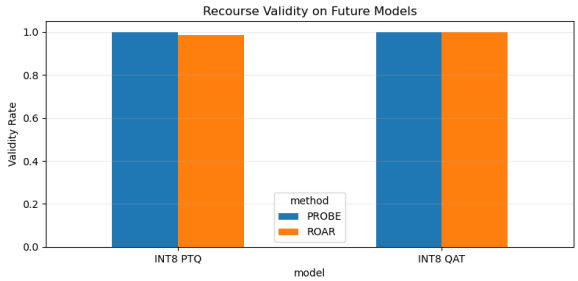

Figure 1: Robust recourse validity on quantized current model (COMPAS)

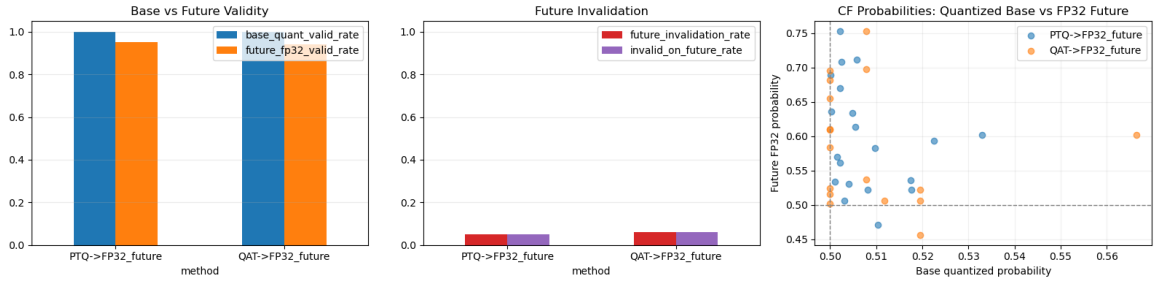

Figure 2: Quantized model generated counterfactual validity on future non-quantized model (GERMAN)

4 Experiments

We conduct three groups of experiments to evaluate the robustness of recourse under model compression and distribution shift. In the first setting, we isolate the effect of model compression while holding the data distribution fixed. A source FP32 model is trained on the Compas [10] dataset and subsequently compressed using INT8 post-training quantization (PTQ) and INT8 quantization-aware training (QAT). Counterfactual robust recourse, using SOTA methods like ROAR and PROBE, is generated for negatively classified instances from the source model (up to 80 samples) and then evaluated on the corresponding compressed models. This setup allows us to measure recourse validity transfer degradation induced purely by quantization, independent of dataset shift. The results are visualized in Figure 1.

In the second and third settings, we now take a look at the robustness of the recourse offered by a standard cost minimizing method like Wachter [12] if the source model is the one that is quantized.

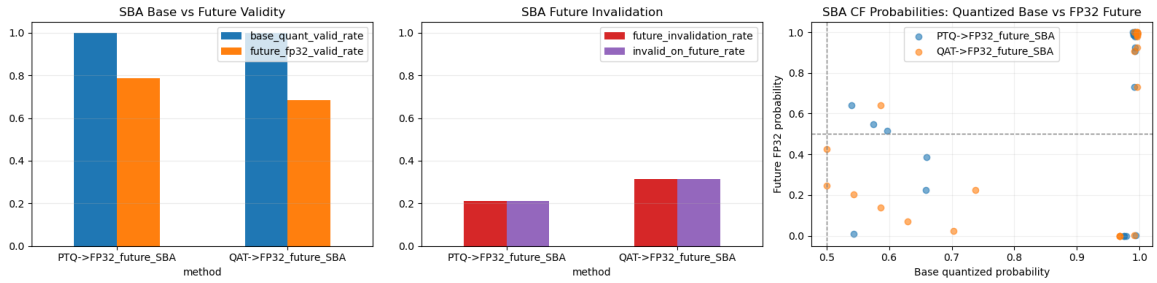

Figure 3: Quantized model generated counterfactual validity on future non-quantized model (SBA)

The difference setting will make use of two different groups of data, with each representing a different kind of shift/change. The German Credit [5] and Corrected German Credit [4] represent the correction data shift. The SBA [7] represents the temporal data shift. In the case of SBA, the current dataset is created by filtering out certain "future" samples from the original data, while the future dataset comprises all available samples.

Recourse is generated using the quantized base models (PTQ and QAT) and subsequently evaluated on the future FP32 model. However, there is a caveat with the optimization style of the Wachter method. Since the method requires a differentiable objective, which requires the derivatives of the model output with respect to the input, we must use the base FP32 as a surrogate for the purpose of optimization. Our counterfactual must still be predicted as the positive class by our quantized model to be considered successful.

This setting captures recourse invalidation arising from both model compression and changes in the underlying data distribution. Results for the German and SBA datasets are shown in Figures 2 and 3, respectively.

5 Discussion

Quantization achieves substantial compression, reducing model size by 50–75% while preserving over 95% of baseline accuracy, indicating that global decision boundaries remain largely intact. In our discussion of counterfactual robustness, we will compare with Table 1 in Appendix A, taken from [11].

When transferring counterfactuals from quantized to future non-quantized models, an unexpected baseline of robustness emerges across both post-training quantization (PTQ) and quantization-aware training (QAT) models. This deviates from the typical boundary-hugging fragility of Wachter's optimization, as seen in Table 1, and likely stems from our required use of a surrogate. Because quantized weights yield negligibly small derivatives [1], we employed a dual-model approach: the base FP32 model guided gradient calculations, while the quantized model strictly evaluated the 0.5 decision threshold. This decoupling inadvertently seems to forces the optimization further along the FP32 loss surface to satisfy the quantized boundary, creating an accidental buffer that enhances survival rates on future models. Building on this surrogate-induced baseline, a distinct dynamic remains between the two quantization methods. Because QAT better preserves boundary fidelity, its counterfactuals stay relatively closer to the edge. Conversely, PTQ introduces significant clipping and rounding errors that arbitrarily distort decision boundaries [2]. This PTQ-induced distortion forces the surrogate-guided search to find even more distant solutions, amplifying the buffer and resulting in higher apparent robustness than QAT. However, it is critical to emphasize that this amplified PTQ robustness is a byproduct of stochastic volatility rather than a reliable feature, making it equally capable of producing highly brittle recourse.

6 Conclusion

In conclusion, our investigation highlights the precarious nature of algorithmic recourse when subjected to the pressures of model updates and compression. While underlying data distribution shifts remain the primary cause for counterfactual invalidation, the quantization process introduces highly nuanced reliability challenges. Specifically, our findings demonstrate that generating recourse on quantized models may require surrogate-guided optimization, which inadvertently creates a buffer that inflates

apparent robustness on future models. Furthermore, while Quantization-Aware Training (QAT) preserves boundary fidelity, Post-Training Quantization (PTQ) induces arbitrary boundary distortions that masquerade as robustness but are fundamentally volatile and unreliable. Consequently, future research must prioritize the development of recourse generation methods intrinsically resilient to both real-world data drift and precision degradation. Establishing adaptive frameworks capable of verifying and updating explanations as models evolve, alongside investigating the recourse implications of other compression paradigms like pruning and knowledge distillation.

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A Appendix

			Correction Shift			Temporal Shift			Geospatial Shift		
Model	Cost	Recourse	Avg Cost	$\mathcal{M}_1$ Validity	$\mathcal{M}_2$ Validity	Avg Cost	$\mathcal{M}_1$ Validity	$\mathcal{M}_2$ Validity	Avg Cost	$\mathcal{M}_1$ Validity	$\mathcal{M}_2$ Validity
LR	L1	CFE	1.02 ± 0.18	1.00 ± 0.00	0.54 ± 0.27	3.57 ± 1.14	1.00 ± 0.00	0.31 ± 0.09	8.37 ± 0.73	0.98 ± 0.03	0.29 ± 0.09
		AR	0.85 ± 0.14	1.00 ± 0.00	0.53 ± 0.21	1.50 ± 0.28	1.00 ± 0.00	0.16 ± 0.06	5.29 ± 0.28	1.00 ± 0.00	0.43 ± 0.14
		ROAR	3.13 ± 0.32	1.00 ± 0.00	0.94 ± 0.08	3.14 ± 0.25	0.99 ± 0.01	$\mathbf{0.98 \pm 0.02}$	10.88 ± 1.67	1.00 ± 0.00	$\mathbf{0.67 \pm 0.19}$
		MINT	4.73 ± 1.56	1.00 ± 0.00	0.93 ± 0.07	NA	NA	NA	NA	NA	NA
		ROAR-MINT	6.77 ± 0.35	1.00 ± 0.00	$\mathbf{1.00 \pm 0.00}$	NA	NA	NA	NA	NA	NA
	PFC	CFE	0.03 ± 0.02	1.00 ± 0.00	0.56 ± 0.33	0.24 ± 0.09	1.00 ± 0.00	0.26 ± 0.11	0.34 ± 0.04	1.00 ± 0.00	0.18 ± 0.10
		AR	0.09 ± 0.02	1.00 ± 0.00	0.54 ± 0.27	0.11 ± 0.02	1.00 ± 0.00	0.09 ± 0.05	0.32 ± 0.03	1.00 ± 0.00	0.24 ± 0.11
		ROAR	0.36 ± 0.08	1.00 ± 0.00	$\mathbf{1.00 \pm 0.00}$	0.44 ± 0.12	0.99 ± 0.01	$\mathbf{0.98 \pm 0.01}$	1.20 ± 0.10	1.00 ± 0.00	$\mathbf{0.91 \pm 0.07}$
		MINT	1.00 ± 1.15	1.00 ± 0.00	0.95 ± 0.08	NA	NA	NA	NA	NA	NA
		ROAR-MINT	1.23 ± 0.05	1.00 ± 0.00	1.00 ± 0.00	NA	NA	NA	NA	NA	NA
NN	L1	CFE	0.55 ± 0.10	1.00 ± 0.00	0.47 ± 0.06	3.78 ± 0.68	1.00 ± 0.00	0.52 ± 0.09	10.09 ± 0.71	1.00 ± 0.00	0.48 ± 0.09
		AR-LIME	0.38 ± 0.15	0.16 ± 0.10	0.31 ± 0.06	1.39 ± 0.13	0.59 ± 0.11	0.65 ± 0.17	9.02 ± 1.57	0.76 ± 0.06	0.83 ± 0.10
		ROAR-LIME	1.83 ± 0.19	0.78 ± 0.06	0.72 ± 0.10	4.90 ± 0.24	0.98 ± 0.02	$\mathbf{0.97 \pm 0.02}$	21.05 ± 3.58	1.00 ± 0.00	$\mathbf{0.97 \pm 0.03}$
		MINT	2.24 ± 1.25	0.81 ± 0.02	0.63 ± 0.11	NA	NA	NA	NA	NA	NA
		ROAR-MINT	8.59 ± 1.70	0.90 ± 0.03	$\mathbf{0.84 \pm 0.04}$	NA	NA	NA	NA	NA	NA
	PFC	CFE	0.06 ± 0.02	1.00 ± 0.00	0.51 ± 0.12	0.19 ± 0.06	1.00 ± 0.00	0.50 ± 0.13	0.48 ± 0.06	1.00 ± 0.00	0.30 ± 0.14
		AR-LIME	0.06 ± 0.03	0.49 ± 0.11	0.56 ± 0.15	0.11 ± 0.01	0.54 ± 0.08	0.62 ± 0.12	0.78 ± 0.15	0.84 ± 0.06	0.82 ± 0.11
		ROAR-LIME	0.64 ± 0.08	0.85 ± 0.07	$\mathbf{0.82 \pm 0.05}$	0.37 ± 0.07	0.99 ± 0.01	$\mathbf{0.99 \pm 0.0}$	1.66 ± 0.21	1.00 ± 0.00	$\mathbf{0.97 \pm 0.04}$
		MINT	0.60 ± 0.16	0.82 ± 0.07	0.64 ± 0.15	NA	NA	NA	NA	NA	NA
		ROAR-MINT	0.60 ± 0.07	0.91 ± 0.04	0.81 ± 0.04	NA	NA	NA	NA	NA	NA

Table 1: Average cost, M1 (original) validity, and M2 (shifted model) validity of recourses across datasets [11]. Here are comparing our own results with the papers results for Model = NN, cost = L1, recourse = CFE (Wachter), and datasets used are Correction and Temporal shifts.